

Tutorial 04/02

1. Before treatment with a new drug 11 people with sleep problems have a median sleeping time of 2 hours per night. A drug is administered and it is known for good scientific reasons that if it has an effect it will increase sleeping time, but some doctors doubt if it will have any effect. Are their doubts justified if the hours per night slept by these individuals after taking the drug are:

3.1 1.8 2.7 2.4 2.9 0.2 3.7 5.1 8.3 2.1 2.4

- (a) Test this without an assumption of symmetry.
 - (b) What would be a more appropriate test with an assumption of symmetry? Carry out this latter test.
2. An instructor wants to use two exams in her classes next year. This year, she gives both exams to the students. She wants to know if the exams are equally difficult by looking at the scores.

Student	Exam 1 Score	Exam 2 Score
Bob	63	69
Nina	65	66
Tim	56	62
Kate	100	91
Alonzo	88	78
Jose	83	87
Nikhil	77	79
Julia	92	88
Tohru	90	85
Michael	84	92

- (a) State the null and alternative hypotheses
 - (b) Carry out an appropriate parametric test stating all assumptions.
 - (c) How would you do a permutation based test for this hypothesis?
 - (d) Find the value of the test statistic and the two sided critical region at 5% level.
3. In a comparison of the visual acuity of deaf and hearing children, eye movement rates are taken on 10 deaf and 10 hearing children. The data are shown in the accompanying table. A clinical psychologist believes that deaf children have greater visual acuity than hearing children. (The larger a child's eye movement rate, the more visual acuity the child possesses.)

Deaf Children	2.75 1.95 3.14 2.17 3.23 2.45 2.30 1.83 2.64 2.23
Hearing Children	1.15 1.23 1.65 2.03 1.43 1.64 1.83 1.96 1.75 1.37

 Carry out an appropriate non-parametric test to evaluate the psychologist's claim at $\alpha=5\%$.